

Campaign B0: Exact Class-A Domain-Wall Calibration

Automated deterministic campaign report¹

¹*Repository experiment review*

(Dated: August 16, 2026)

This report records the fixed-rank CPU benchmark generated by campaign 20260816_191957. It calibrates the analysis pipeline and is not evidence about the adaptive circuit. The transverse gate status is **PASS**; the declared analysis width is $N_x = 20$.

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I. PROTOCOL AND PROVENANCE

Both the coupled spatial-mass interface and the hard-exterior analogue use the canonical `classA.U1FGTN._domain_wall_hamilton` source and exactly $N_x N_y$ occupied orbitals. The locked configuration and complete source hashes are in `manifest.json`. The mass criterion $m/(2\pi|v|/N_y) \leq 0.1$ is necessary but not sufficient: the accepted width and the next larger scanned width must both pass every single-geometry calibration at $N_y = 48$.

II. CORRECTION AND LEGACY RECONCILIATION

This version supersedes campaign 20260816_184251 for interpretation without modifying that run. Version 1 injected a normalized packet at the center of the retained half-cylinder and reduced its spreading to an absolute center of mass. That observable cancelled counterpropagating modular components. It also chose $N_x = 12$ from the hybridization mass alone, although the coupled wall correlator and the modular endpoint diagnostic do not converge until $N_x = 20$. The discrepancy is therefore classified as an estimator mismatch compounded by finite-width underconvergence, not as a failure of modular charge spreading.

The corrected construction follows the legacy exact-domain-wall notebook: it reduces a translated half-cylinder, injects at both entanglement endpoints, and retains the complete $N(x, y, t)$ evolution. Unlike the aggregate legacy movies, the four wall/endpoint sources are evolved separately. Their symmetry-cancelling signed statistic is

$$D_w(t) = \frac{1}{2} [\bar{y}_{w,0}(t) + \bar{y}_{w,L-1}(t) - (L-1)], \quad L = N_y/2. \quad (1)$$

Its slope is a modular-time handedness diagnostic; its magnitude is not compared with a physical or spectral velocity.

III. REPRESENTATIVE RESULTS

IV. ENTROPY ACCOUNTING

The full-width strip has two entanglement boundaries and intersects both physical domain walls. Each three-column physical-wall contour window includes both entanglement cuts. Those window sums are diagnostic contributions to the full entropy, not isolated subsystem entropies. The saved products test equality of the contour sum and the entropy, equality of the two wall slopes, and the factor-of-two relation between a wall-window slope and the full-strip slope.

TABLE I. Accepted-width representative at $N_y = 48$.

construction	$ v $	m	width ratio	c_1	β	$v_{\text{mod}}^{0,1}$	wall flows
coupled	0.983856	2.1e-09	1.63e-08	1.0001	2.0745	-2.93, +4.536	-1, +1
hard_exterior	0.983856	1.94e-09	1.51e-08	1.0001	2.0744	-2.6, +4.406	-1, +1

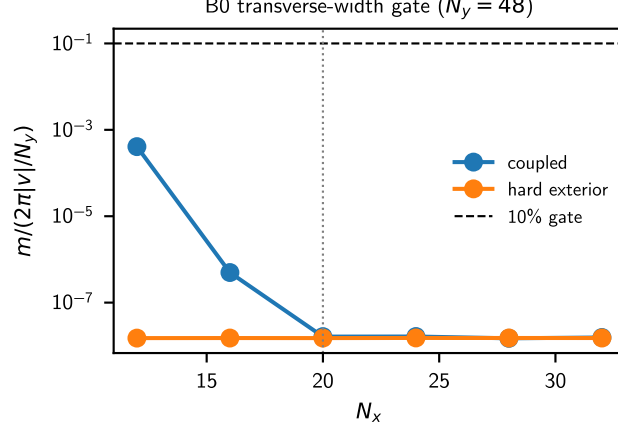


FIG. 1. The hybridization-mass condition remains a necessary transverse-width gate.

V. ACCEPTANCE LEDGER

requirement	status	details
small-system formula and gauge preflight	PASS	completed before production stages
fixed-rank half filling	PASS	every geometry must contain exactly $N_x N_y$ occupied orbitals
Hermiticity, projector, and contour numerics	PASS	absolute tolerance 1e-10
consecutive full-calibration transverse-width gate	PASS	selected $N_x=20$; mass threshold 0.1; selected and next larger widths must pass every single-geometry gate
coupled: opposite, compatible spectral velocities	PASS	$v_0=-0.990726$, $v_1=0.990726$, relative mismatch=0
coupled: finite transverse localization lengths	PASS	$\xi_0=0.29557$, $\xi_1=0.29557$
coupled: Renyi $q=1,2,3$ full-strip $c=1$	PASS	$q=1$: $c=1.0001$, $q=2$: $c=0.99591$, $q=3$: $c=0.99336$
coupled: single-physical-wall contour factor of two	PASS	relative errors=[8.468181767162797e-05, 8.468181752530057e-05]; each window includes both entanglement cuts
coupled: wall squared-correlator exponent two	PASS	$\beta_{\text{etas}}=[2.0744675989239347, 2.0744675989889787]$; all four candidate models remain in correlator_fits.csv
coupled: physical response chirality	PASS	response velocities=[-0.8720279720279717, 0.8720279720279717]; spectral velocities=[-0.9907263837597539, 0.9907263837597539]; relative magnitude errors=[0.11980947886068002, 0.11980947886068002]
coupled: response linearity and charge conservation	PASS	charge drift=2.21e-16; epsilon relative error=5e-07
coupled: stable opposite endpoint modular velocities	PASS	primary velocities=[-2.930447164248109, 4.535630720918731]; sign stable=[True, True]; minimum retention=0.3245; norm drift=3.33e-15
coupled: nonsingular overlap tracking	PASS	minimum singular value=0.999904
coupled: unit wall flow and zero net flow	PASS	signed flows=[-1, 1]
hard_exterior: opposite, compatible spectral velocities	PASS	$v_0=-0.990726$, $v_1=0.990726$, relative mismatch=0
hard_exterior: finite transverse localization lengths	PASS	$\xi_0=0.222754$, $\xi_1=0.222754$

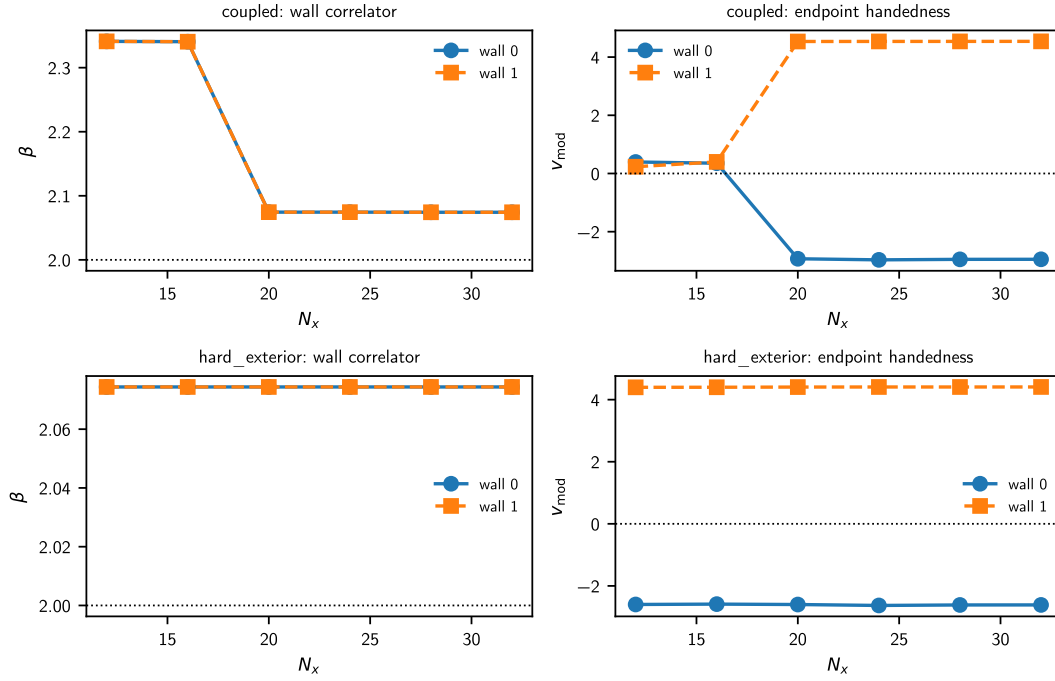


FIG. 2. Local-observable convergence. The coupled correlator and endpoint modular handedness exclude $N_x = 12, 16$ even though their fitted hybridization masses pass.

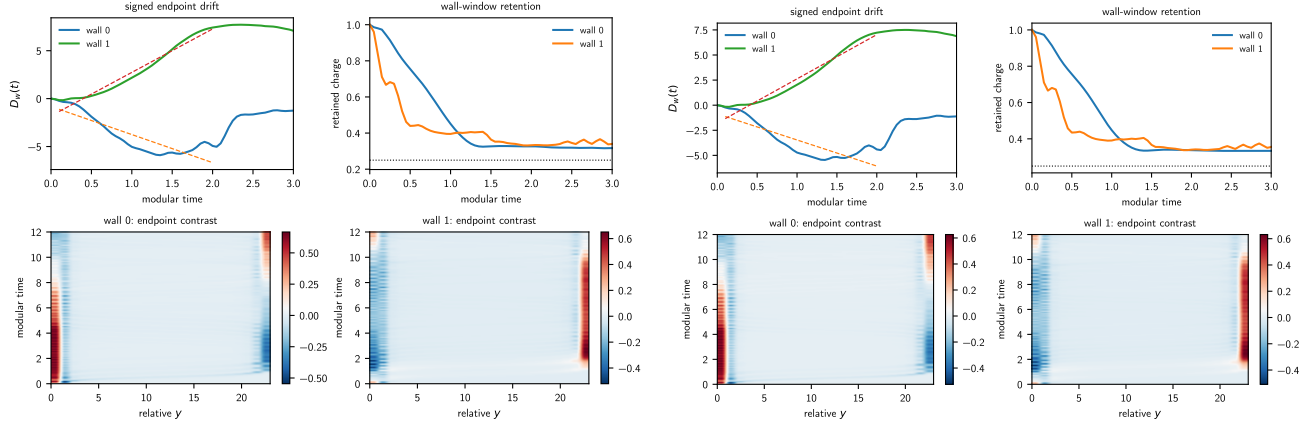


FIG. 3. Separate entanglement-endpoint injections, signed drifts, and wall retention for the accepted-width coupled and hard-exterior constructions.

hard_exterior: Renyi $q=1,2,3$ full-strip $c=1$ PASS
 hard_exterior: single-physical-wall contour fac- PASS
 tor of two

hard_exterior: wall squared-correlator expo- PASS
 nent two
 hard_exterior: physical response chirality PASS

hard_exterior: response linearity and charge PASS
 conservation

$q=1$: $c=1.0001$, $q=2$: $c=0.99591$, $q=3$: $c=0.99332$
 relative errors= $[8.629596877662848e-05, 8.629596836606801e-05]$; each window includes both entanglement cuts
 betas= $[2.074368474806555, 2.074368474815762]$; all four candidate models remain in correlator_fits.csv
 response velocities= $[-0.8720279720279717, 0.8720279720279717]$; spectral velocities= $[-0.9907263837597539, 0.9907263837597539]$; relative magnitude errors= $[0.11980947886068002, 0.11980947886068002]$
 charge drift= $3.8e-16$; epsilon relative error= $5e-07$

hard_exterior: stable opposite endpoint modu- PASS lar velocities	primary velocities=[-2.5998489202228035, 4.405780712633448]; sign stable=[True, True]; mini- mum retention=0.3349; norm drift=4e-15
hard_exterior: nonsingular overlap tracking PASS	minimum singular value=0.999999
hard_exterior: unit wall flow and zero net flow PASS	signed flows=[-1, 1]
33/65/129-point, both-sign, seam-relocation PASS	all validation rows require unit absolute wall flow and
twist validation	seam/uniform spectral agreement

The machine-readable ledger retains the same decisions and their thresholds. Any failed item remains explicit rather than changing the estimator or gate.